

# Propositional Logic



# This Lecture

In the last lecture we introduced logic formulas.

In this lecture we are going to use logic to derive things.

The two important components in this lecture are:

- Conditional Statements
- Arguments

# Conditional Statement (Implication)

If p then q

$$p \longrightarrow q$$

p implies q

p is called the **hypothesis**; q is called the **conclusion**

**antecedent**

**consequent**

The department says: "If your GPA is 4.0, then you will have full scholarship."

When is the above sentence false?

- It is false when your GPA is 4.0 but you don't receive full scholarship.
- But it is not false if your GPA is below 4.0.
  - So, if your GPA is below 4.0, it is true irrespective of whether you receive full scholarship or not.

Another example: "If there is a match of Bangladesh, then there is no class."

When is the above sentence false?

# Logic Operator

$\rightarrow ::= \text{IMPLIES}$

| $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

**Convention:** if we don't say anything wrong, then it is not false, and thus true.

An implication is **true** provided  $P$  is false or  $Q$  is true (or both), and **false** otherwise.

# Example Implications

- The most common type of statement in mathematics is the implication.
- Consider the *Pythagorean Theorem*:

$$a^2 + b^2 = c^2$$

Would you agree that this is the theorem?

Is this a statement? **NO**

- The equation is true as long as  $a$  and  $b$  are the legs of a right triangle and  $c$  is the hypotenuse.
- We thus make it a statement:  
*If  $a$  and  $b$  are the legs of a right triangle with hypotenuse  $c$ , then  $a^2 + b^2 = c^2$*

# Implications

- This is a reasonable way to think about implications:
  - our claim is that the conclusion (“then” part) is true, but on the assumption that the hypothesis (“if” part) is true.
  - We make no claim about the conclusion in situations when the hypothesis is false.
- This is why when the hypothesis is false, we don’t say anything wrong, i.e., our statement is NOT false, and thus true.

# Implication

- What do you mean if you say,  $p \rightarrow q$  is false?
- You mean:
  - $p$  is true and
  - $q$  is false

# Implication Exercise

Which of the following are True/False?

- If  $1 = 1$ , then most horses have 4 legs. **TRUE**
- If  $0 = 1$ , then  $1 = 1$ . **TRUE**
- If 8 is a prime number, then the 7624th digit of  $\pi$  is an 8. **TRUE**
- If the 7624th digit of  $\pi$  is an 8, then  $2 + 2 = 4$ .

**TRUE**

An implication is *true* provided  $P$  is false or  $Q$  is true (or both), and *false* otherwise.



# Logical Equivalence

$$p \rightarrow q \equiv ?$$

We want a  
(compound)  
statement  
equivalent to  
left hand side

If you see a question in the above form,  
there are usually 3 ways to deal with it.

- (1) Truth table
- (2) Use logical rules
- (3) Intuition

## If-Then as Or

$$p \rightarrow q \equiv ?$$

An implication is **true** provided  $P$  is false or  $Q$  is true (or both), and **false** otherwise.

| $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

Idea 2: Look at the false rows, **negate** and take the **"and"**.

$$\begin{aligned} & \neg(P \wedge \neg Q) \\ \equiv & \neg P \vee Q \end{aligned}$$

DeMorgan's Laws

- If you don't give me all your money, then I will kill you.
- Either you give me all your money or I will kill you (or both).
- If you talk to her, then you can never talk to me.
- Either you don't talk to her or you can never talk to me (or both).

## If-Then as Or

$$P \rightarrow Q \equiv \sim P \text{ or } Q$$

P

Q

- If you don't give me all your money, then I will kill you.
- Either you give me all your money or I will kill you (or both).

$\sim P$

P

Q

- If you talk to her, then you can never talk to me.
- Either you don't talk to her or you can never talk to me (or both).

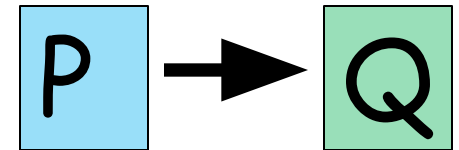
$\sim P$

# Direct proofs of implications

- To prove an implication  $P \rightarrow Q$ :
  - it is enough to assume  $P$ , and from it, deduce  $Q$ .
  - i.e., you must explain why  $Q$  is true, but you *get to* assume  $P$  is true first.
- Direct proof is the easiest style of proof and has the advantage that such a proof often does a great job of explaining *why* the statement is true.

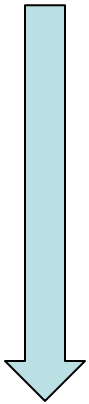
# Direct proofs of implications

- If two numbers  $a$  and  $b$  are even, then their sum  $a + b$  is even.



Proof:

assume  $P$



Deduce  $Q$

- Suppose the numbers  $a$  and  $b$  are even.
- This means that  $a = 2k$  and  $b = 2j$  for some integers  $k$  and  $j$ .
- The sum is then  $a + b = 2k + 2j = 2(k + j)$ .
- Since  $k + j$  is an integer, this means that  $a + b$  is even.

QED

## Negation of If-Then

$$\neg(p \rightarrow q) \equiv ?$$

$$\neg(\text{if } p \text{ then } q) \equiv$$



$$p \vee q \quad (\text{if } \neg p \text{ then } q)$$



$$p \equiv \neg q \quad (\text{if } \neg p \text{ then } \neg q)$$



$$\neg p \vee \neg q \equiv (\text{if } p \text{ then } \neg q)$$



$$p \rightarrow q \equiv \neg p \vee q$$

## Negation of If-Then

$$\neg(p \rightarrow q) \equiv ?$$

$$\neg(P \rightarrow Q)$$

$$\equiv \neg(\neg P \vee Q)$$

previous slide

$$\equiv \neg\neg P \wedge \neg Q$$

DeMorgan

$$\equiv P \wedge \neg Q$$

## Negation of If-Then

$$\neg(p \rightarrow q) \equiv P \wedge \neg Q$$

P

Q

- If you eat an apple everyday, then you have no toothache.
- You eat an apple everyday but you have toothache.

$\sim Q$

negate

P

Q

- If my computer is not working, then I cannot finish my homework.
- My computer is not working but I can finish my homework.

$\sim Q$

negate



# Contrapositive

The **contrapositive** of "if  $p$  then  $q$ " is "if  $\sim q$  then  $\sim p$ ".

**Statement:** If you are a CS year 1 student,  
then you are taking CSC 2110.

**Contrapositive:** If you are not taking CSC 2110,  
then you are not a CS year 1 student.

**Statement:** If  $x^2$  is an even number,  
then  $x$  is an even number.

**Contrapositive:** If  $x$  is an odd number,  
then  $x^2$  is an odd number.

**Fact:** A conditional statement is logically equivalent to its contrapositive.

Often it is easier to analyze the contrapositive.

# Proofs

**Statement:** If  $P$ , then  $Q$

**Contrapositive:** If  $\neg Q$ , then  $\neg P$ .

Proof using the Truth table

| $P$ | $Q$ | $P \rightarrow Q$ | $\neg Q$ | $\neg P$ | $\neg Q \rightarrow \neg P$ |
|-----|-----|-------------------|----------|----------|-----------------------------|
| T   | T   | T                 | F        | F        | T                           |
| T   | F   | F                 | T        | F        | F                           |
| F   | T   | T                 | F        | T        | T                           |
| F   | F   | T                 | T        | T        | T                           |

Proof using logical rules...

$$P \rightarrow Q \equiv \neg P \vee Q \equiv Q \vee \neg P \equiv \neg Q \rightarrow \neg P$$

# Inverse

- The **Inverse** of an implication  $P \rightarrow Q$  is the implication  $\sim P \rightarrow \sim Q$  .
- The inverse is NOT logically equivalent to the original implication.
  - That is, whether the inverse of an implication is true is independent of the truth of the implication

# Converse

- The **converse** of an implication  $P \rightarrow Q$  is the implication  $Q \rightarrow P$ .
- The converse is NOT logically equivalent to the original implication.
  - That is, whether the converse of an implication is true is independent of the truth of the implication

# Converse and Inverse

- Converse and Inverse of a statement are logically equivalent to each other...
- Why/How?
  - $P \rightarrow Q$
  - Inverse:  $\sim P \rightarrow \sim Q$
  - Converse:  $Q \rightarrow P$
  - Contrapositive of  $\sim P \rightarrow \sim Q$  is  $\sim\sim Q \rightarrow \sim\sim P$ , i.e.,  $Q \rightarrow P$ .

# Converse

- Mathematics is overflowing with examples of true implications which have a false converse.
  - If a number greater than 2 is prime, then that number is odd.
    - CONVERSE: If a number ( $>2$ ) is odd, then it is a prime
    - However, just because a number is odd does not mean it is prime.
  - If a shape is a square, then it is a rectangle.
    - CONVERSE: If a shape is a rectangle, then it is a square
    - But it is false that if a shape is a rectangle, then it is a square.

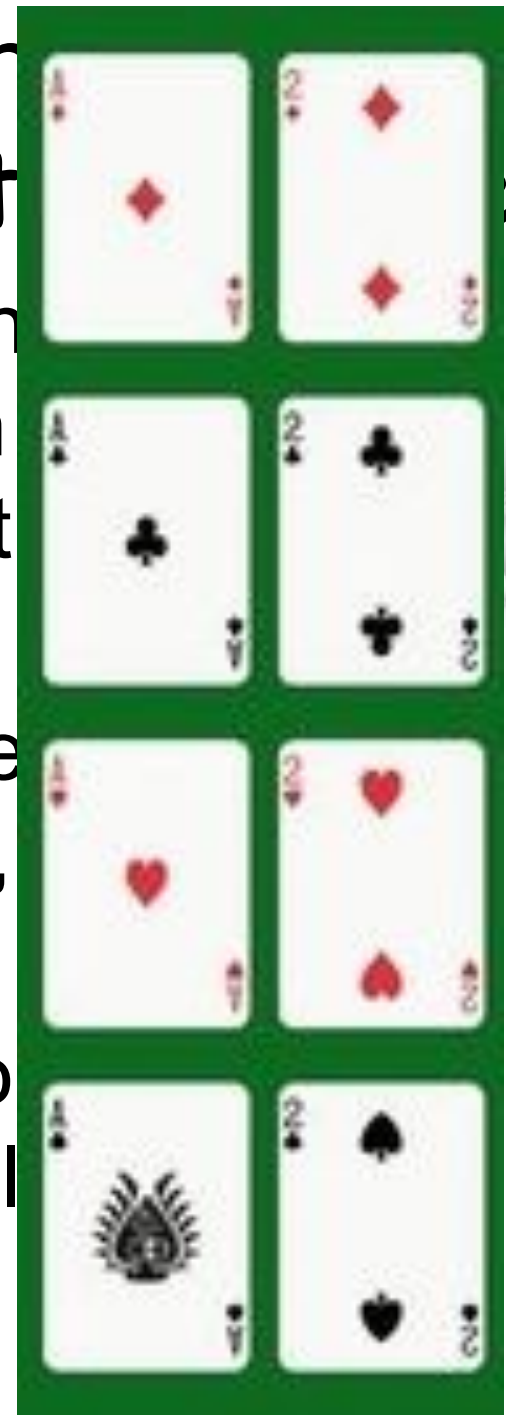
# Converse

- Sometimes the converse of a true statement is also true.
- For example, the Pythagorean theorem has a true converse: if  $a^2 + b^2 = c^2$ , then the triangle with sides  $a$ ,  $b$ , and  $c$  is a *right* triangle.
- Whenever you encounter an implication in mathematics, it is always reasonable to ask whether the converse is true.

When both  $P \rightarrow Q$  and  $Q \rightarrow P$  are true, we say that  $P$  and  $Q$  are equivalent and write  $P \leftrightarrow Q$ . This is also called the biconditional. We will come back to this soon....

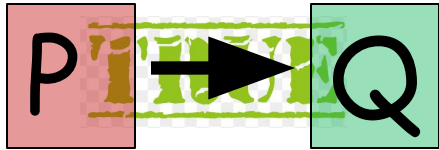
# Implication and Contradiction

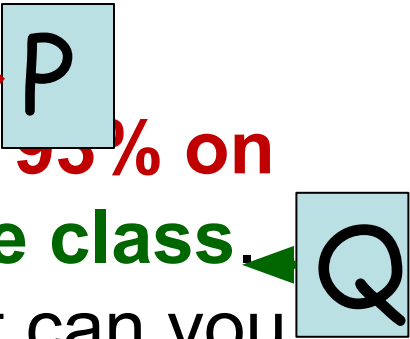
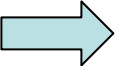
- True or False: If you draw any nine cards from a regular deck, then you must have at least three cards all of the same suit.
- contrapositive: If you *don't* have three cards all of the same suit, you don't have nine cards.
- True: you can at most have two cards of each of the four suits, for a total of eight cards (or fewer).





# Understanding implications



- Suppose I tell Rimpi that **if she gets a 93% on her final**, then **she will get an A in the class**.  
Assuming that what I said is true, what can you conclude in the following cases:  

- Rimpi gets a 93% on her final.   $P$  is true; since  $P \rightarrow Q$  is true;  $Q$  must be true
  - Rimpi gets an A.
- Rimpi gets an A in the class.   $Q$  is true;  $P \rightarrow Q$  is given true; But  $P$  still can be both true or false.
  - Cannot conclude anything.
- Rimpi does not get a 93% on her final.   $P$  is false;  $P \rightarrow Q$  is (automatically) true;  $Q$  can be both true or false.
  - Cannot conclude anything.
- Rimpi does not get an A in the class.   $Q$  is false;  $P \rightarrow Q$  is given true;  $P$  then must be false.
  - Rimpi does not get a 93% on her final.

# Implies, if, only if

| $P$ | $Q$ | $P \rightarrow Q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

Look at the True rows;  
When  $P$  is true,  
what about  $Q$ ?  $Q$  is also true!  
So,  $P$  is sufficient to make  $Q$  true  
(Note that  $Q$  can still be true  
when  $P$  is not.)

- We can say:  $Q$  if  $P$  (i.e., if  $P$  then  $Q$ )
  - $P$  is a sufficient condition for  $Q$
- Or we can say  $P$  only if  $Q$ .
  - i.e., if not  $q$ , then not  $p$ .
    - (i.e., contrapositively: if  $p$  then  $q$ .)
  - $Q$  is a necessary condition for  $P$

Look at the True rows;  
Can  $P$  be true without  $Q$  being true?  
No! So,  $Q$  is necessary for  $P$  to be true.

If you work hard., you will succeed

$S \rightarrow R$

## If, Only-If

R = you will  
succeed

S = you work  
hard

- You will succeed **if** you work hard.
- You will succeed **only if** you work hard.

R if S means "if S then R" or equivalently "S implies R"

We also say S is a **sufficient condition** for R.

R only if S means "if R then S" or equivalently "R implies S"

We also say S is a **necessary condition** for R.

You will succeed **if and only if** you work hard.

P if and only if (iff) Q means P and Q are logically equivalent.

That is, P implies Q and Q implies P.

# Math vs English

**Parent:** if you don't clean your room, then you can't watch a DVD.

$C$

$D$

This sentence says  $\neg C \rightarrow \neg D$   $\xrightarrow{\text{contrapositive}}$   $D \rightarrow C$

In real life it also means  $C \rightarrow D$  

So  $C \leftrightarrow D$  

**Mathematician:** if a number  $x$  greater than 2 is not an odd number,  
then  $x$  is not a prime number.

This sentence says  $\neg O \rightarrow \neg P$

But of course it doesn't mean  $O \rightarrow P$

$O = x > 2$  is an odd  
number

$P = x > 2$  is a prime  
number

# Necessary, Sufficient Condition

**Mathematician:** if a number  $x$  greater than 2 is not an odd number, then  $x$  is not a prime number.

$O = x > 2$  is an odd number

$P = x > 2$  is a prime number

This sentence says  $\neg O \rightarrow \neg P$

But of course it doesn't mean  $O \rightarrow P$

Contrapositive

**Mathematician:** if a number  $x$  greater than 2 is a prime number then  $x$  is an odd number.

Being an odd number  $> 2$  is a **necessary condition** for this number to be prime.  
 $P$  only if  $O$

THIS WAS MISTAKENLY DISCUSSED AS  
"O only if P" IN THE LECTURE

Being a prime number  $> 2$  is a **sufficient condition** for this number to be odd.  
 $O$  if  $P$

## Necessary AND Sufficient Condition

$$\longleftrightarrow ::= \text{IFF}$$

| $P$ | $Q$ | $P \longleftrightarrow Q$ |
|-----|-----|---------------------------|
| T   | T   | T                         |
| T   | F   | F                         |
| F   | T   | F                         |
| F   | F   | T                         |

**Note:**  $P \longleftrightarrow Q$  is equivalent to  $(P \longrightarrow Q) \wedge (Q \longrightarrow P)$

**Note:**  $P \longleftrightarrow Q$  is equivalent to  $(P \longrightarrow Q) \wedge (\neg P \longrightarrow \neg Q)$

Is the statement "x is an even number if and only if  $x^2$  is an even number" true?

# Another Proof!

- $x$  is an even number if and only if  $x^2$  is an even number.
- $p = x$  is an even number
- $q = x^2$  is an even number
- So we need to prove:  $p \leftrightarrow q$ 
  - i.e.,  $p \rightarrow q$  and  $q \rightarrow p$

$x$  is an even number if and only if  $x^2$  is an even number

Proof:

$p = x$  is an even number  
 $q = x^2$  is an even number

a.  $p \rightarrow q$

$x$  is even  $\Rightarrow x = 2k$

$\therefore x^2 = (2k)^2 = 4k^2$ , which is even

b.  $q \rightarrow p$

~~$x^2$  is even  $\Rightarrow x^2 = 2k$~~

~~$\therefore x = \sqrt{2k} =$~~  

b.  $q \rightarrow p \equiv \neg p \rightarrow \neg q$  CONTRAPOSITIVE

$x$  is NOT even  $\Rightarrow x = 2k + 1$

$\therefore x^2 = (2k + 1)^2 = 4k^2 + 2k + 1$ , which is NOT even



# Hypothesis and Conclusion

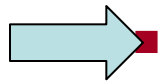
May not be related

- If  $1 = -1$ , then I am the King of the world
- If computers are machines, then Mashrafi is a Cricket Player
- If  $2+2 = 5$ , then Chandler is a girl.
- But for meaningful statements, we formulate an implication when there is a relation between the hypothesis and conclusion.

# This Lecture

## ■ Conditional Statements

- The meaning of IF and its logical forms
- Contrapositive
- If, only if, if and only if

 Arguments

# Argument Form

An argument is a sequence of statements **forms**.

All statement **forms** but the final one are called **assumptions** or **hypothesis** or **premises**.

The final statement **form** is called the **conclusion**.

An argument **form** is **valid** if:

No matter what particular statements are substituted for the statement variables in its premises, whenever all the assumptions are true, then the conclusion is true.

# Argument

An argument is a sequence of statements.

All statements but the final one are called **assumptions** or **hypothesis**.

The final statement is called the **conclusion**.

An argument is **valid** if: its form is valid

If today is Wednesday, then yesterday is Tuesday.

Today is Wednesday.

∴ Yesterday is Tuesday.

This is valid, because its form is valid: See next slide...

# Modus Ponens

If  $p$  then  $q$ .  
 $p$   
 $\therefore q$

If typhoon, then class cancelled.  
Typhoon.  
 $\therefore$  Class cancelled.

| assumptions |     |                   | conclusion |     |
|-------------|-----|-------------------|------------|-----|
| $p$         | $q$ | $p \rightarrow q$ | $p$        | $q$ |
| T           | T   | T                 | T          | T   |
| T           | F   | F                 | T          | F   |
| F           | T   | T                 | F          | T   |
| F           | F   | T                 | F          | F   |

Modus ponens is Latin meaning "method of affirming".

# Modus Tollens

If  $p$  then  $q$ .  
 $\sim q$   
 $\therefore \sim p$

If typhoon, then class cancelled.  
Class not cancelled.  
 $\therefore$  No typhoon.

| assumptions |     |                   | conclusion |          |
|-------------|-----|-------------------|------------|----------|
| $p$         | $q$ | $p \rightarrow q$ | $\sim q$   | $\sim p$ |
| T           | T   | T                 | F          | F        |
| T           | F   | F                 | T          | F        |
| F           | T   | T                 | F          | T        |
| F           | F   | T                 | T          | T        |

Modus tollens is Latin meaning "method of denying".

# Equivalence

A student is trying to prove that propositions  $P$ ,  $Q$ , and  $R$  are all true. She proceeds as follows.

First, she proves three facts:

- $P$  implies  $Q$
- $Q$  implies  $R$
- $R$  implies  $P$ .

Then she concludes,

``Thus  $P$ ,  $Q$ , and  $R$  are all true.''

Proposed argument:

$$\frac{(P \rightarrow Q), (Q \rightarrow R), (R \rightarrow P)}{P \wedge Q \wedge R}$$

assumption

Is it valid?

conclusion

# Valid Argument?

$$(P \rightarrow Q), (Q \rightarrow R), (R \rightarrow P)$$

Is it valid?

$$P \wedge Q \wedge R$$

assumptions

conclusion

| P | Q | R |
|---|---|---|
| T | T | T |
| T | T | F |
| T | F | T |
| T | F | F |
| F | T | T |
| F | T | F |
| F | F | T |
| F | F | F |

| $P \rightarrow Q$ | $Q \rightarrow R$ | $R \rightarrow P$ |
|-------------------|-------------------|-------------------|
| T                 | T                 | T                 |
| T                 | F                 | T                 |
| F                 | T                 | T                 |
| F                 | T                 | T                 |
| T                 | T                 | F                 |
| T                 | F                 | T                 |
| T                 | T                 | F                 |
| T                 | T                 | T                 |

| $P \wedge Q \wedge R$ | OK? |
|-----------------------|-----|
| T                     | yes |
| F                     | yes |
| F                     | yes |
| F                     | yes |
| F                     | yes |
| F                     | yes |
| F                     | yes |
| F                     | no  |

Assumptions are true, but not the conclusion.

To prove an argument is not valid, we just need to find a counterexample.



# Valid Arguments?

If p then q.  
q  
 $\therefore$  p



If you are a fish, then you drink water.  
You drink water.  
You are a fish.

| assumptions |   |                   | conclusion |   |
|-------------|---|-------------------|------------|---|
| p           | q | $p \rightarrow q$ | q          | p |
| T           | T | T                 | T          | T |
| T           | F | F                 | F          | T |
| F           | T | T                 | T          | F |
| F           | F | T                 | F          | F |

Assumptions are true, but not the conclusion.

**NOT VALID**

# Valid Arguments?

| assumptions |   |                   |          | conclusion |
|-------------|---|-------------------|----------|------------|
| p           | q | $p \rightarrow q$ | $\sim p$ | $\sim q$   |
| T           | T | T                 | F        | F          |
| T           | F | F                 | F        | T          |
| F           | T | T                 | T        | F          |
| F           | F | T                 | T        | T          |

Assumptions are true, but not the conclusion.

If p then q.  
 $\sim p$   
 $\therefore \sim q$

If you are a fish, then you drink water.  
You are not a fish.  
You do not drink water.

**NOT VALID**

## Exercises

$$\begin{array}{c} p \\ \therefore p \vee q \end{array}$$

$$\begin{array}{c} p \\ \therefore p \wedge q \end{array}$$

$$\begin{array}{c} p \wedge q \\ \therefore p \end{array}$$

$$\begin{array}{c} p \vee q \\ \therefore p \end{array}$$

$$\begin{array}{c} p \vee q \\ \neg q \\ \therefore p \end{array}$$

$$\begin{array}{c} p \longrightarrow q \\ q \longrightarrow r \\ \therefore p \longrightarrow r \end{array}$$

## Exercises

$$p$$

**VALID**

$$\therefore p \vee q$$

| P | Q | P or Q |
|---|---|--------|
| 0 | 0 | 0      |
| 0 | 1 | 1      |
| 1 | 0 | 1      |
| 1 | 1 | 1      |

$$p \wedge q$$

**VALID**

$$\therefore p$$

| P | Q | P and Q | P |
|---|---|---------|---|
| 0 | 0 | 0       | 0 |
| 0 | 1 | 0       | 0 |
| 1 | 0 | 0       | 1 |
| 1 | 1 | 1       | 1 |

$$p \vee q$$

$$\neg q$$

**VALID**

$$\therefore p$$

| P | Q | P or Q | $\neg Q$ | P |
|---|---|--------|----------|---|
| 0 | 0 | 0      | 1        | 0 |
| 0 | 1 | 1      | 0        | 0 |
| 1 | 0 | 1      | 1        | 1 |
| 1 | 1 | 1      | 0        | 1 |

| P | Q | P and Q |
|---|---|---------|
| 0 | 0 | 0       |
| 0 | 1 | 0       |
| 1 | 0 | 0       |
| 1 | 1 | 1       |



$$p$$

$$\therefore p \wedge q$$

| P | Q | P or Q | P |
|---|---|--------|---|
| 0 | 0 | 0      | 0 |
| 0 | 1 | 1      | 0 |
| 1 | 0 | 1      | 1 |
| 1 | 1 | 1      | 1 |



$$p \vee q$$

$$\therefore p$$

| P | Q | R | $P \rightarrow Q$ | $Q \rightarrow R$ | $P \rightarrow R$ |
|---|---|---|-------------------|-------------------|-------------------|
| 0 | 0 | 0 | 1                 | 1                 | 1                 |
| 0 | 0 | 1 | 1                 | 1                 | 1                 |
| 0 | 1 | 0 | 1                 | 0                 | 1                 |
| 0 | 1 | 1 | 1                 | 1                 | 1                 |
| 1 | 0 | 0 | 0                 | 1                 | 0                 |
| 1 | 0 | 1 | 0                 | 1                 | 1                 |
| 1 | 1 | 0 | 1                 | 0                 | 0                 |
| 1 | 1 | 1 | 1                 | 1                 | 1                 |



$$p \rightarrow q$$

$$q \rightarrow r$$

$$\therefore p \rightarrow r$$

## More Exercises

$$\begin{array}{l} \neg p \longrightarrow q \\ \neg q \\ \therefore p \end{array}$$

$$\begin{array}{l} \neg p \longrightarrow \neg q \\ \therefore p \longrightarrow q \end{array}$$

$$\begin{array}{l} \neg p \longrightarrow \neg q \\ \therefore q \longrightarrow p \end{array}$$

## More Exercises

$$\neg p \rightarrow q$$

$$\neg q$$

$$\therefore p$$

| P | ~P | Q | $\neg p \rightarrow q$ | ~Q | P |
|---|----|---|------------------------|----|---|
| 0 | 1  | 0 | 0                      | 1  | 0 |
| 0 | 1  | 1 | 1                      | 0  | 0 |
| 1 | 0  | 0 | 1                      | 1  | 1 |
| 1 | 0  | 1 | 1                      | 0  | 1 |

| P | Q | ~P | ~Q | $\neg p \rightarrow \neg q$ | $p \rightarrow q$ |
|---|---|----|----|-----------------------------|-------------------|
| 0 | 0 | 1  | 1  | 1                           | 1                 |
| 0 | 1 | 1  | 0  | 0                           | 1                 |
| 1 | 0 | 0  | 1  | 1                           | 0                 |
| 1 | 1 | 0  | 0  | 1                           | 1                 |

$$\neg p \rightarrow \neg q$$

$$\therefore p \rightarrow q$$

$$\neg p \rightarrow \neg q$$

$$\therefore q \rightarrow p$$

| P | Q | ~P | ~Q | $\neg p \rightarrow \neg q$ | $q \rightarrow p$ |
|---|---|----|----|-----------------------------|-------------------|
| 0 | 0 | 1  | 1  | 1                           | 1                 |
| 0 | 1 | 1  | 0  | 0                           | 0                 |
| 1 | 0 | 0  | 1  | 1                           | 1                 |
| 1 | 1 | 0  | 0  | 1                           | 1                 |

Valid argument  $\nrightarrow$  True conclusion

True conclusion  $\nrightarrow$  Valid argument

Valid argument  $\rightarrow$  True conclusion



If John Lennon was a Rock Star, then John Lennon had red hair.  
John Lennon was a Rock Star.  
 $\therefore$  John Lennon had red hair



modus ponens

But Conclusion is False!!!

True conclusion  $\rightarrow$  Valid argument



If New York is a big city, then New York has tall buildings.  
New York has tall buildings.  
 $\therefore$  New York is a big city.

But Conclusion is True!!!



$$1 = -1$$

$\therefore$  Today is Tuesday.



Which form is this?

| P | Q |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |

$P \equiv 1 = -1$   
 $\therefore Q \equiv \text{Today is Tuesday.}$



| P | $\sim P$ | C | $P \wedge C$ | P |
|---|----------|---|--------------|---|
| 0 | 1        | 0 | 0            | 0 |
| 1 | 0        | 0 | 1            | 1 |

**Contradiction**

$$\neg p \rightarrow C$$

Here C means Contradiction

$$\therefore p$$

If you can show that the assumption that the statement p is false leads logically to a contradiction, then you can conclude that p is true.

This is similar to the method of denying (modus tollens)

# Knights and Knaves

Knights always tell the truth.  
Knaves always lie.

A says: B is a knight.  
B says: A and I are of opposite type.

Suppose A is a knight.  
Then B is a knight (because what A says is true).  
Then A is a knave (because what B says is true)  
A contradiction.

So A must be a knave.

So B must be a knave (because what A says is false).

# Complex Deductions

You are about to leave for BUET in the morning and discover that you don't have your glasses. You know that the following statements are true:

- A. If my glasses are on the dining table, then I saw them at breakfast.
- B. I was reading the newspaper in the living room or I was reading the newspaper in the dining room.
- C. If I was reading the newspaper in the living room, then the glasses are on the coffee table.
- D. I did not see my glasses at breakfast
- E. If I was reading Sohel sir's lecture in bed, then my glasses are on the bed side table
- F. If I was reading the newspaper in the dining room, then my glasses are on the dining table.

**The glasses are on the coffee table.**

# Complex Deductions

|                     |                |                |
|---------------------|----------------|----------------|
| $p \rightarrow q$   | a. $p \vee q$  | b. $p \vee q$  |
| $\sim q$            | $\sim q$       | $\sim p$       |
| $\therefore \sim p$ | $\therefore p$ | $\therefore q$ |

You are about to leave for BUET in the morning and discover that you don't have your glasses. You know that the following statements are true:

- A. If my glasses are on the dining table, then I saw them at breakfast.
- B. I was reading the newspaper in the living room or I was reading the newspaper in the dining room.
- C. If I was reading the newspaper in the living room, then the glasses are on the coffee table.
- D. I did not see my glasses at breakfast
- E. If I was reading Sohel sir's lecture in bed, then my glasses are on the bed side table
- F. If I was reading the newspaper in the dining room, then my glasses are on the dining table.

By A and D and modus tollens:

- 1. My glasses are not on the dining table

By F and 1 and modus tollens:

- 2. I was not reading the newspaper in the dining room

By B and 2 and elimination:

- 3. I was reading the newspaper in the living room

By C and 3 and modus ponens:

- 4. The glasses are on the coffee table.

$$\begin{array}{l}
 p \rightarrow q \\
 p \\
 \therefore q
 \end{array}$$

The glasses are on the coffee table.

Table 1.3.1 Valid Argument Forms

|                       |                                                      |                                     |                                                                        |                                                    |
|-----------------------|------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------|----------------------------------------------------|
| <b>Modus Ponens</b>   | $p \rightarrow q$<br>$p$<br>$\therefore q$           | <b>Elimination</b>                  | <b>a.</b> $p \vee q$<br>$\sim q$<br>$\therefore p$                     | <b>b.</b> $p \vee q$<br>$\sim p$<br>$\therefore q$ |
| <b>Modus Tollens</b>  | $p \rightarrow q$<br>$\sim q$<br>$\therefore \sim p$ | <b>Transitivity</b>                 | $p \rightarrow q$<br>$q \rightarrow r$<br>$\therefore p \rightarrow r$ |                                                    |
| <b>Generalization</b> | <b>a.</b> $p$<br>$\therefore p \vee q$               | <b>Proof by Division into Cases</b> | $p \vee q$<br>$p \rightarrow r$<br>$q \rightarrow r$<br>$\therefore r$ |                                                    |
| <b>Specialization</b> | <b>b.</b> $q$<br>$\therefore p \vee q$               |                                     |                                                                        |                                                    |
|                       | <b>a.</b> $p \wedge q$<br>$\therefore p$             |                                     |                                                                        |                                                    |
|                       | <b>b.</b> $p \wedge q$<br>$\therefore q$             |                                     |                                                                        |                                                    |
| <b>Conjunction</b>    | $p$<br>$q$<br>$\therefore p \wedge q$                | <b>Contradiction Rule</b>           | $\sim p \rightarrow c$<br>$\therefore p$                               |                                                    |

# Quick Summary

## ■ Conditional Statements

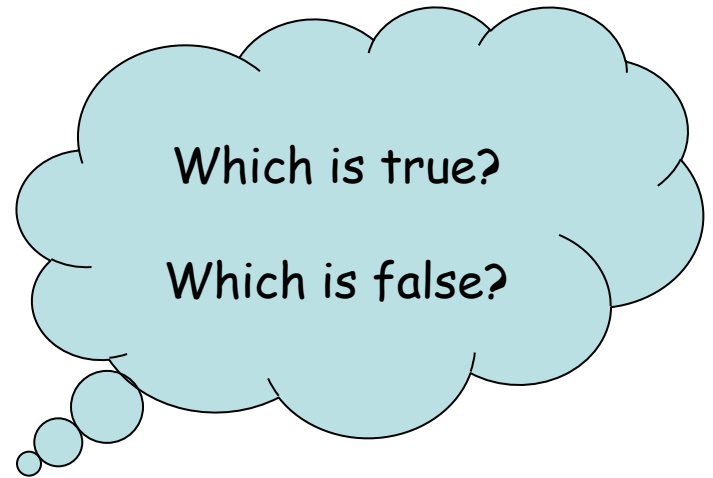
- The meaning of IF and its logical forms
- Contrapositive
- If, only if, if and only if

## ■ Arguments

- definition of a valid argument
- method of affirming, denying, contradiction

Key points:

- (1) Make sure you understand conditional statements and contrapositive.
- (2) Make sure you can check whether an argument is valid.



Which is true?

Which is false?

"The sentence below is **false**."

"The sentence above is **true**."